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AI-Driven Smart Agriculture: An Integrated Approach for Soil Analysis, Irrigation, and Crop-Fertilizer Recommendations

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ABSTRACT Modern agriculture faces critical challenges, including climate change induced water shortage, inefficient resource utilization, and reducing soil health which reduce sustainability and reduce crop yields. Traditional farming methods, dependent on manual observation and generalized practices, often lack the precision needed to address these growing issues. To meet these demands, this study presents an artificial intelligence (AI) powered smart agriculture system that integrates deep learning (DL), Internet of Things (IoT), and explainable artificial intelligence (XAI) to enable intelligent, data-driven farming. This study addresses key challenges such as ineffective soil management, poor irrigation practices, and the absence of personalized crop and fertilizer recommendations. IoT-based sensors provide real-time soil analysis, measuring critical parameters such as nutrient levels, moisture, and temperature. Advanced DL models were employed for various tasks in the AI-Driven Smart Agriculture System. The Transformer-based Tabular Learning (TTL) model for irrigation advice gives an accuracy of 99.13%. For soil analysis, the TabNet regressor achieved a root mean square error (RMSE) of 1.51 and an R^2 score of 98.7%, while XAI techniques were used to explain and interpret its predictions. The Sparse Weighted Fusion Transformer (SwiFT) model for crop recommendation gives an accuracy of 98.75%. For fertilizer recommendation, the TabNet classifier achieved an accuracy of 99.3%, with XAI techniques applied to explain the reasoning behind its predictions. These results demonstrate that the system delivers precise and actionable insights. A weather Application Programming Interface (API) is integrated to support timely irrigation decisions through accurate forecasting. A chatbot interface is also added which enhances system usability, accessibility and user-friendliness for farmers. The proposed model improves crop productivity, optimizes resource use, and supports adaptation to environmental changes. By leveraging AI and IoT technologies, the proposed model gives a solution which helps in sustainable agriculture, environmental resilience, and global food security, thus benefiting stakeholders ranging from individual farmers to large-scale agricultural business. Future work will explore real-time adaptive control mechanisms and expand model generalization across diverse geographical region.

INDEX TERMS XAI, DL, TabNet, SwiFT, RMSE, IOT sensors, Weather Forecasting, Smart Agriculture, Crop Recommendation, Fertilizer Recommendation.

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I. INTRODUCTION

While agriculture has shaped human civilization for centuries, it has to now support roughly 8 billion people around the world. Farmers today are struggling with serious challenges like changing weather, water shortages, and poor soil quality, all of which are making it harder to grow healthy crops. Traditional farming approaches, which often relied on generational wisdom and physical labor, are no longer sufficient to address the needs of an ever-increasing population and its unstable environmental conditions [1].

Motivation for this research arises from an immediate need to modernize agriculture through advance technologies that enhance productivity while promoting sustainability. Current systems for irrigation and crop recommendations rely on fixed rules or basic machine learning (ML), which often fail to understand the complex relationships and real-world challenges [2]. Over 70% of the world's freshwater is used for farming, but a large portion of it is wasted due to poor and unplanned irrigation practices. [3]. Small-scale farmers often have a lack of affordable access to soil health diagnosis and precise weather forecasting, which hinders their ability to make informed decisions. These gaps underline the requirement for a scalable, explainable and cost-effective solution ensuring that innovative agricultural solutions are available and effective for all types of farming communities [4].

Combining the strengths of both tabular and transformer-based models provides actionable insights [5]. Transformer models, in particular, give an accurate and most prominent results. Even with all the advances in technology, significant gaps and challenges remain unsolved [6]. Globally, farming consumes over 70% of our freshwater, much of which is wasted due to poor irrigation methods. Smallholder farmers-who produce around 34% of the world's food, often lack access to essential tools like soil testing and reliable weather forecasts, making it harder for them to manage resources efficiently [7].

The key contributions of paper are summarized as follows:

- Transformer-based tabular learning (TTL): A novel tabular learning approach is introduced to detect complex relationships within sensor data. It enables precise and adaptive irrigation recommendations that enhance efficiency and sustainability.
- Sparse Weighted fusion transformer (SwiFT): A smart and lightweight model is developed to detect even subtle patterns in farm data, enabling accurate and confident crop recommendations.
- TabNet with explainable AI (XAI): TabNet, combined with XAI, is used for soil health analysis by highlighting key feature importance. This facilitates interpretable fertilizer recommendations to support informed decision-making.
- IoT Sensor and Weather API Integration: Real-time soil and environmental data are combined with accurate weather forecasts to provide timely and personalized farming advice, helping reduce resource waste and boost productivity.

By integrating the IoT sensors for soil and environmental monitoring shown in Figure 1, the system collects real-time data. The weather API provides detailed forecasts, while TTL analyzes the combined data to generate personalized irrigation recommendations. TabNet with XAI helps to analyze soil conditions and recommend suitable fertilizers. While SwiFT effectively identifies the most suitable crops based on detailed patterns in agricultural data.

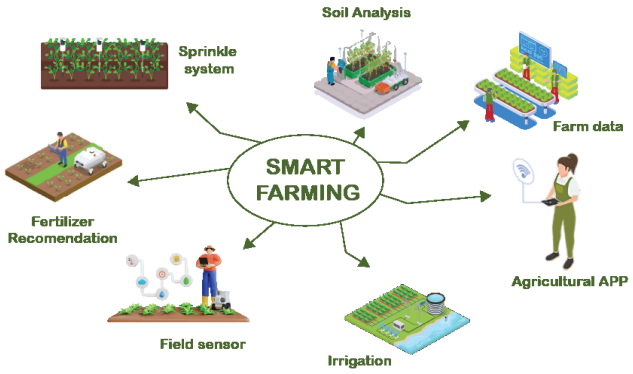

Figure 1. Smart Farming (soil analysis, irrigation system, fertilizer recommendation

The scope of this study involves the development and validation of an integrated, data-driven framework for smart agriculture. It focuses on structured tabular data from agricultural environments, emphasizing real-time efficiency, interpretability, and adaptability to various crop yields and farm sizes[8]. The system is designed to handle diverse crops, soil types, and weather conditions, making it adaptable to a wide range of farming scenarios [9]. Although the current implementation relies on publicly available datasets and simulated sensor inputs, the architecture is intended for practical deployment in IoT-enabled farming systems bridging the gap between advanced research and real-world agricultural applications [10].

A 7-in-1 multifunctional soil sensor provided in Figure 2 was used to collect essential real-time field data for smart agriculture applications. This sensor accurately measures soil moisture, temperature, pH, and macronutrient levels (nitrogen, phosphorus, and potassium), which are critical inputs for our AI-driven models. Built with corrosion-resistant metal probes and sealed with high-density epoxy resin. It ensures durability and long-term performance, even when buried in soil or exposed to moisture. The sensor's IP68 rating makes it waterproof and suitable for continuous use in harsh outdoor environments. Making ideal for data collection in agriculture, plant cultivation, and greenhouse monitoring. The high-quality data from this device served as a reliable foundation for crop prediction, irrigation advice, and fertilizer recommendations in our system.

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Figure 2. **7-in-1 multifunctional soil sensor**

This paper is structured as follows: Section II summarizes related work on recommendation framework for a smart agriculture system. Section III demonstrates the conceptual architecture of the proposed framework and describes the process details. Section IV presents the experimental results and analysis. Finally, the conclusions of this study are summarized in Section V.

II. RELATED WORK

This section explores how AI and IoT-enabled agricultural systems contribute to improved soil health and water management, while offering enhanced crop and fertilizer recommendations to support sustainable farming. TABLE 1 summarizes the research gaps identified from the existing state-of-the-art studies.

A. STATE OF ART WORKS

Recent advances in smart agriculture have leveraged the synergy of IoT technologies and ML models to enhance sustainable food production [11][12][13]. TABLE 1 summarizes the key research gaps identified from the current literature. Earlier systems relied on simple rule-based controls combined with wireless sensors for basic monitoring. However, these systems lacked the ability to learn or adapt, making them less effective under changing field conditions [14]. For crop recommendation, traditional classifiers such as Decision Trees and Naïve Bayes have been widely adopted, typically achieving moderate accuracy 85% to 88% [15]. Models such as SVR and linear regression have achieved up to 93% accuracy for soil and fertilizer recommendations. However, these models are computationally intensive and not well-suited for capturing complex, non-linear patterns commonly found in agricultural data. [16].

Ref [17] reviewed recent advancements in crop prediction using deep learning (DL) methods, including Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), and Long Short-Term Memory (LSTM) models, along with hybrid architectures applied in agricultural contexts. Models such as ANN and random forest regression have been employed for tasks like irrigation scheduling, crop recommendation, and fertilizer management, achieving accuracies above 90% [18]. ML algorithms such as decision trees, and SVM have also been applied to classify soil types using cone penetration test (CPT) data [19]. The study found that CNNs outperformed Evolutionary Neural Networks (ENNs), achieving approximately 87% accuracy.

Ensemble approaches, such as combining random forest and SVM models, have achieved commendable accuracy rates—reaching up to 94% in crop and soil analysis. However, these methods often increase system complexity, posing implementation challenges. Gradient Boosting Regression has demonstrated exceptional performance, with 99.3% accuracy in fertilizer recommendation tasks. Yet, its high computational requirements limit its practicality for large-scale or resource-constrained applications [20]. Despite these technological advances, several challenges persist. Rule-based systems are inherently incapable of learning from data, and traditional machine learning models struggle to model intricate, non-linear relationships among features. Neural networks, while powerful, frequently suffer from computational overhead and the risk of overfitting. Moreover, many current solutions lack transparency and are difficult to operationalize [21]. These limitations highlight the necessity for scalable, interpretable, and integrated frameworks capable of processing heterogeneous agricultural data and delivering actionable insights for sustainable agricultural practices [22].

B. PROBLEM DESCRIPTION

Most existing approaches focus on isolated components of the agricultural ecosystem such as climate, soil, or crop yield resulting in segmented and incomplete decision support [23]. The complex, high-dimensional nature of agricultural data further complicates processing and analysis. Traditional ML and DL models often struggle to scale effectively or perform optimally under these conditions [24]. Additionally, many state-of-the-art models require extensive training data and high computational resources, limiting their widespread adoption, especially in resource-constrained environments [25].

An experiment using four machine learning models and a focused training approach was conducted to predict crop yield. Among these, the Random Forest algorithm outperformed others, achieving 91.5% accuracy [26]. Temperature and rainfall were identified as the most influential factors affecting crop yield. ML algorithms have demonstrated superior performance over traditional statistical methods by offering deeper insights into the factors influencing agricultural productivity [27]. These models serve as valuable tools for policymakers, supporting informed decisions in agricultural planning and resource allocation. The study's approach can be applied to other regions with similar climatic conditions and agricultural practices [28]. A key strength of this work lies in its detailed data preprocessing techniques; however, a more thorough discussion on data quality and its impact on model performance would further enhance the study's robustness [29].

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TABLE 1. Research gap summary from existing works on smart agriculture system.

Ref.	Smart Agriculture	ML/ DL	Constraints	Findings	Research Gap
[30]	Irrigation Advice via IoT Sensor	Rule-based threshold control (WSN-based system)	Temperature, ambient humidity	Qualitative validation	No learning capability.
[31]	Crop Recommendation	Decision Tree classifier	Soil color, moisture content	Accuracy: 85%	Doesn't model the feature interactions well
[32]	Crop Recommendation	Naïve base Classifier	Moisture, humidity	Accuracy: 88%	Strong independence assumption
[33]	Soil Analysis via IoT Devices	Support Vector Regression (SVR)	Soil color, moisture content, composition	Accuracy: 93%; MSE: 0.25	High computational cost
[34]	Fertilizer Recommendation	Linear Regression model	Nutrition (sodium, potassium, nitrogen)	$R^2 = 0.85$	Can't model nonlinear relationships
[35]	Irrigation Advice & Crop Recommendation	Artificial Neural Network (ANN)	Rainfall distribution, moisture	Accuracy: 91%	Prone to overfitting if not regularized
[36]	Soil Analysis & Fertilizer Recommendation	Random Forest Regression	Soil color, temporal patterns	Accuracy: 95%; MSE: 0.15	Hard to interpret individual decisions without XAI
[37]	Crop Recommendation	Support Vector Machine (SVM) classifier	Soil Nutritions	Accuracy: 92%	Needs careful parameter tuning
[38]	Irrigation Advice	Convolutional Neural Network (CNN)–based control system	Ground water, Rainfall distribution	Accuracy: ~90%	CNNs are image-focused
[39]	Crop Recommendation & Soil Analysis	Ensemble Model (Random Forest + SVM)	Spatial and temporal patterns	Accuracy: 94%	Complex implementation
[40]	Fertilizer Recommendation	Gradient Boosting Regression	Crop demage, area, production	Accuracy: 99.3	High training time

SVM combined with ensemble techniques, have been used to analyze soil series data, resulting in improved crop predictions. Water management is especially critical in agriculture, particularly in regions with limited water supply [41]. Recent advances in AI and deep learning have led to smarter irrigation scheduling systems. These systems help conserve water by accurately predicting when and how much to irrigate. One effective approach integrates ML with predictive control techniques to create intelligent, data-driven irrigation plans [42]. Clustering approaches, such as k-means, divide regions into different management zones for irrigation based on topography and soil features. Subsequently, LSTM networks are used for modeling soil moisture dynamics [43].

This study proposes an integrated, XAI-based smart agriculture framework that unifies climate, soil, and crop yield data. The proposed model delivers interpretable and actionable recommendations for sustainable crop production by combining advanced models—such as SwiFT, TTL, and TabNet enhanced with XAI—with real-time IoT sensor data and weather APIs. This approach addresses the limitations of traditional methods and is designed to handle the complexity and scale of agricultural data. The system supports transparent decision-making and enables practical deployment across diverse farming environments. This represents a novel and more effective solution, as summarized in Table 2

Table 2 Task-wise Models and Addressed Limitations

Task Category	Algorithm/technique	Limitations Addressed
Crop Recommendation	Sparse Weighted Fusion Transformer	Don't overfit and simple, straightforward implementation
Fertilizer Recommendation	Tabular Network and Explainable Artificial Intelligence	Model the feature interactions well
Soil Analysis via IoT Devices	TabNet+XAI	Low computational cost
Irrigation Advice via IoT Sensors	Transformer-based Tabular Learning	High Learning Capability

III. METHODOLOGY

Agriculture is transitioning from traditional practices to smart, data-driven decisions, powered by AI to enhance accuracy and efficiency in irrigation advice and other recommendations. The proposed model effectively handles unseen data, with data acquisition and preprocessing steps illustrated in Figure 3.

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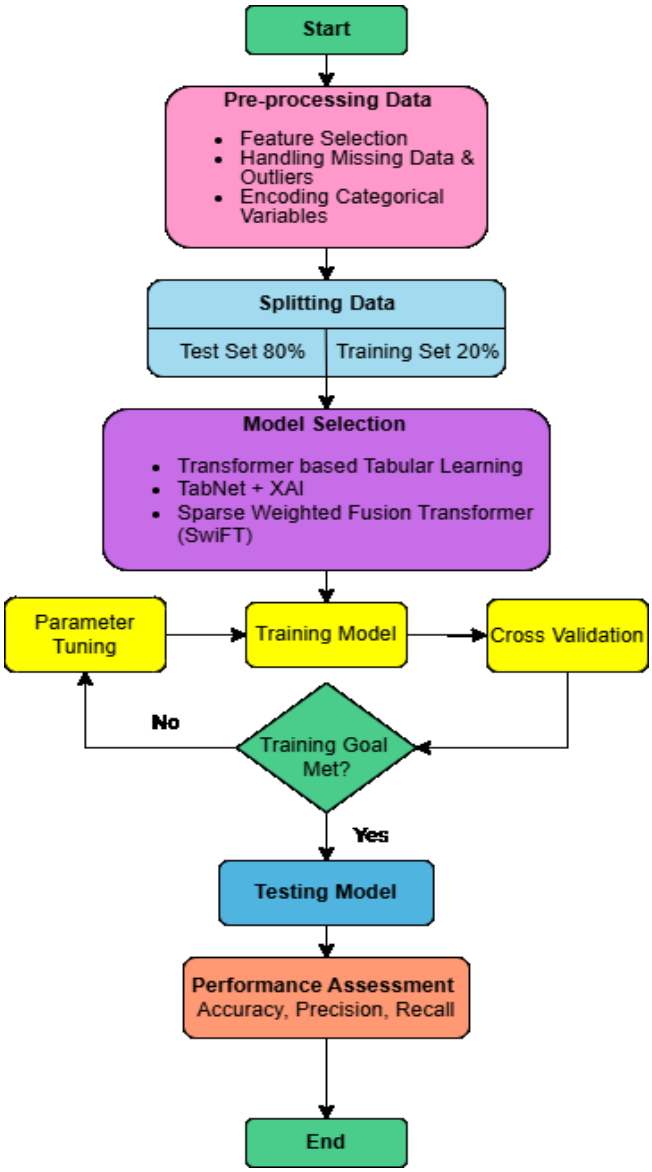

Figure 3. **Use of the proposed model on unseen agricultural data**

This study presents an AI-driven framework for smart agriculture, where the Transformer-based TTL model provides adaptive irrigation recommendations. TabNet, integrated with Explainable AI (XAI), offers clear insights for soil analysis and fertilizer selection, while the SwiFT model delivers accurate crop recommendations. The initial stage of the framework involves diverse data acquisition methods, utilizing multiple datasets to ensure robust model training and performance.

A. DATA ACQUISITION AND PREPROCESSING

The initial phase involves comprehensive data collection from diverse sources, including soil nutrient profiles, real-time environmental sensor readings historical weather data, and crop yield records. Data preprocessing steps include normalization, handling of missing values, and encoding techniques to ensure compatibility with DL models. This multi-modal dataset provides a robust foundation for capturing the complex interactions present within the agricultural ecosystem. Data pre-processing includes following steps stated below:

- The first step is **data cleaning**, which includes identifying and correcting missing or incomplete values. In real-world agricultural data, missing entries are common due to sensor malfunction or inconsistent recordings. These gaps are filled using statistical methods such as mean or median imputation, depending on the nature of the feature. Additionally, **normalization** is applied to scale numerical values like temperature, moisture, and nutrient levels to a consistent range—typically between 0 and 1. This ensures that no single feature dominates model learning due to its scale.
- Another key step is **outlier removal**. Extreme or unusual values in the dataset caused by faulty sensor readings or rare environmental conditions are identified using standard deviation thresholds or visual analysis techniques. These outliers are either corrected if possible or removed to prevent them from skewing model predictions.
- Since the dataset also includes categorical information, such as crop labels (e.g., rice, wheat, maize), **encoding techniques** are used to convert these categories into numerical format. This is achieved using **label encoding** or **one-hot encoding**, making the data compatible with machine learning algorithms, which require numeric input.

B. SPARSE WEIGHTED FUSION TRANSFORMATION (SwiFT) FOR CROP RECOMMENDATION

SwiFT is a neural network architecture designed to efficiently integrate information from multiple data sources. It was chosen for its effectiveness in managing complex agricultural datasets, which involve overlapping variables, dynamic conditions, and deeply connected factors influencing crop performance and resource utilization [44]. Unlike traditional machine learning models such as Random Forests, Support Vector Machines (SVMs), and Multilayer Perceptron (MLPs) SwiFT leverages transformer-based attention mechanisms that can capture long-range dependencies among features.

- **Feature Interdependence and Long-Range Attention**

Agricultural data includes diverse inputs such as weather conditions, soil properties, and farmer activity records. SwiFT's attention mechanism effectively captures the interconnections among these features, outperforming traditional models in understanding long-range dependencies. This results in more accurate and dependable predictions.

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- **Efficient Feature Fusion**

SwiFT incorporates a dedicated feature fusion layer following the self-attention module, which refines and unifies complex information from multiple sources. This design enhances the model's ability to recognize key patterns, leading to improved accuracy and reliability in its predictions.

Algorithm 1 Sparse Weighted Fusion Transformation

Algorithm: SwiFT Predict

Input:

X – input feature matrix, W_Q, W_K, W_V – Learnable weights for query, key, and value matrices, A – Adaptive Sparsity matrix, F_fuse – Fusion function for weights representations, Y_true – Ground truth labels for training (for supervised learning).

Output:

Y_pred – Predicted class or output values

Procedure:

1. Input encoding: Transform input features X into query, key and value representation using:

$$Q = XW_Q, K = XW_K, V = XW_V$$

2. Sparse Attention Computation: Compute sparse attention scores using:

$$A_{sparse} = softmax\left(\frac{QK^T}{\sqrt{d_k}}.A\right)$$

Where A enforces sparsity by making irrelevant connections.

3. Weighted Fusion: Apply weighted fusion to refine feature representations:

$$F_{fuse} = A_{sparse}V$$

4. Transformer Block Processing: pass the fused feature through multiple SwiFT layers, incorporating positional encoding and normalization.
5. Prediction Generation: Apply a Classification or regression head to generate final output:

$$Y_{pred} = \mathrm{MLP}(F_{fuse})$$

6. Loss Calculation (for Training Only): Compute loss using entropy or RMSE, and update weights via backpropagation.
7. Return Prediction: Output Y_{pred}

End Algorithm

C. TRANSFORMER BASED TABULAR LEARNING (TTL) FOR IRRIGATION ADVICE

Traditional models like CNNs are highly effective for image processing because they excel at recognizing local patterns in fixed spatial positions. However, this approach doesn't translate well to agricultural data, which is typically in tabular form and lacks a consistent structure or spatial order. TTL overcomes this limitation by using a self-attention mechanism that captures the relationships among features such as soil moisture, temperature, and humidity. Each feature is first transformed into a rich, high-dimensional representation, which is then passed through multiple transformer encoder layers. These layers enable the model to learn meaningful, context-aware patterns across the data. Unlike CNNs or standard fully connected networks, TTL is well-suited to modeling the complex, non-linear relationships commonly present in agricultural datasets. It demonstrates strong generalization capabilities, adapts to unseen conditions, and is more robust when handling real-world, noisy sensor data. This makes TTL an excellent fit for applications such as soil analysis, irrigation planning, and crop decision-making [45].

Algorithm 2 Transformer-based Tabular Learning

Algorithm: Transformer-base Tabular Classifier

Input:

$$X, W_E, T, W_{fc}, Y$$

Output:

Y_{pred} - Predicted class labels

Procedure:

1. Feature Embedding: convert input features into a higher dimensional representation using linear transformation:

$$E(X) = XW_E + b_E$$

2. Transformer Encoder Processing the embedded features pass through two transformer encoder layers. Each layer contains multi-head self-attention followed by feedforward network (FFN).
   a. Multi-head self-attention:
   Each attention head computes:

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)$$

Multi-head attention output:

$$\mathrm{MultiHead}(X) = \mathrm{Concat}(head_1, ..., head_h)W_o$$

   b. Feedforward Network (FFN): After attention the data is passed through a two-layer feedforward network with ReLU activation:

$$FFN(X) = \sigma(W_1X + b_1)W_2 + b_2$$

3. Classification Layer: After passing through the transformer encoder, the final feature represents is mapped to the output classes:

$$P(Y \mid X) = \mathrm{softmax}(W_{fc}Z + b_{fc})$$

The predicted class is:

$$\hat{Y} = \arg\max P(Y \mid X)$$

4. Training Process:
   a. Loss Function (Cross Entropy): For multiclass classification the loss function computed as:

$$\ell = -\sum_{i=1}^{C} Y_i \log P(Y_i \mid X)$$

   b. Optimization (Adam): Gradient updates are computed using back propagation:

$$\theta \longleftarrow \theta - \eta \nabla_\theta \ell$$

5. Training Loop Equation: For each epoch, update the model parameters:

$$\theta^{(t+1)} = \theta^{(t)} - \eta \sum_{i=1}^{N} \nabla_\theta \ell$$

End Algorithm

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where X is the input tabular data of shape (N, D), W_E is Learning weight matrix for embedding, T is Transformer encoder with self-attention layers, W_{fc} represents Fully connected layer for classification, Y indicate Target labels for supervised learning.

Feature Embedding:

$$E(X) = XW_E + b_E \quad (1)$$

where W_E is a learnable weight matrix for embedding, b_E is a bias term and $E(X)$ the embedded representation of input

Multi-head self-attention:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) \quad (2)$$

where $Q = E(X)W_{QV}K, K = E(X)W_K, V = E(X)W_V$, W_Q, W_K, W_V are learnable weights for query, key, and value d_K is the key dimension

Multi-head attention output:

$$\text{MultiHead}(X) = \text{Concat}(head_1, ..., head_h)W_o \quad (3)$$

· W_o is the output projection matrix

Feedforward Network (FFN):

$$FFN(X) = \sigma(W_1X + b_1)W_2 + b_2 \quad (4)$$

where W_1, W_2 are weight matrices, b_1, b_2 are biases, σ is the ReLU activation function

Classification Layer:

$$P(Y \mid X) = \text{softmax}(W_{fc}Z + b_{fc}) \quad (5)$$

where Z is the output from transform encoder and W_{fc}, b_{fc} are learnable classification parameter

Loss Function (Cross Entropy):

$$\ell \quad \sum_{i=1}^{C} y_i \log P(Y_i \mid X) \quad (6)$$

C is the number of classes and y_i is the true label (hot encoder)

Optimization (Adam):

$$\theta \longleftarrow \theta - \eta \nabla_\theta \ell \quad (7)$$

· η is the learning rate

Training Loop Equation:

$$\theta^{(t+1)} = \theta^{(t)} - \eta \sum_{i=1}^{N} \nabla_\theta \ell \quad (8)$$

η is the learning rate X is the input, and Y Target labels.

D. TABNET COMBINED WITH EXPLAINABLE AI (XAI) FOR SOIL AND FERTILIZER ANALYSIS

TabNet is a deep learning (DL) model specifically designed to handle structured, tabular data, making it ideal for analyzing soil conditions. Unlike traditional models that often treat all input features equally, TabNet learns to focus selectively on the most relevant ones—such as soil moisture, pH, and nutrient levels—at each decision step. When combined with explainable AI (XAI) methods like SHAP or LIME, TabNet not only provides accurate fertilizer or treatment recommendations but also explains which factors influenced those decisions.

Algorithm 3 TabNet + XAI

Algorithm: TabNet_LIME_Predict

Input:

$$x_{\text{new}},\ X_{\text{train}},\ Y_{\text{train}},\ f_{\text{TabNet}},\ \text{LIME}_{\text{explainer}}$$

Output:

$\hat{y}$ - the predict label for x_{new}

$\Phi(x_{\text{new}})$ - feature importance explanation for LIME

Procedure:

1. Preprocessing: Normalize and preprocesses x_{new} based on training data distribution.
2. Prediction using TabNet:

   pass x_{new} through the trained TabNet model:

   $$P(y \mid x_{\text{new}}) - f_{TabNet}(x_{\text{new}})$$

   Retrieve the predicted class label:

   $$\hat{y} \quad \arg\max P(y \mid x_{\text{new}})$$
3. Generate LIME Explanations:

   Create perturbed samples $\overline{X}$ around x_{new} :

   $$\overline{X} - \{x_{\text{new}} + \varepsilon_i \mid \varepsilon_i \sim \text{N}(0, \sigma^2)\}$$

   Compute model predictions on perturbed sample:

   $$P(y \mid \overline{X}) - f_{\text{TabNet}}\left(\overline{X}\right)$$

   Fit a locally weighted linear model g(x) to approximate

   $$f_{\text{TabNet}} : g(x) - \beta_o + \sum_{i-1}^{d} \beta_i x_i$$
4. Extract Explanation: Retrieve feature importance weights

   $$\Phi(x_{\text{new}}) - (\beta o, \beta_1, \ldots, \beta_d)$$
5. Return Prediction and Explanation: Output y along with feature importance $\Phi(x_{\text{new}})$.

End Algorithm

E. INTEGRATION AND WORK FLOW

The system begins by collecting diverse types of data—ranging from a farmer's day-to-day knowledge to real-time sensor readings from the soil, as well as historical farming records. This information is then preprocessed to ensure consistency and reliability before being passed through intelligent AI models that generate accurate predictions, actionable recommendations, and data-driven decisions. These models analyze soil health to suggest optimal irrigation timings, recommend the most suitable crops for current conditions, and identify the best fertilizers to use. Meanwhile, the system continuously monitors performance, adjusts predictions as necessary, and updates the models as new data becomes available. All of these valuable insights

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are shared with farmers, researchers, and policymakers to support more informed decision-making and promote efficient, sustainable farming practices. The methodology of the proposed model is illustrated in Figure 4.

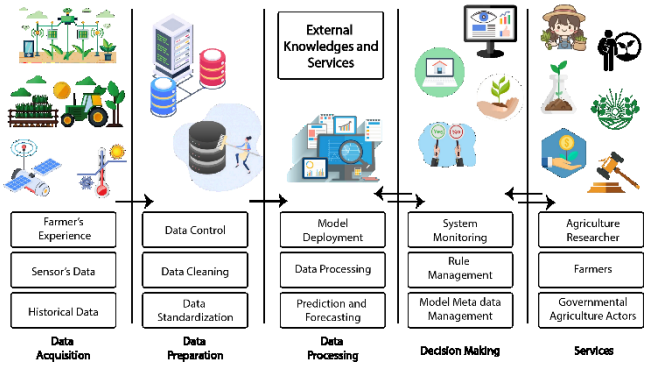

Figure 4. Methodology of Proposed model

IV. RESULTS AND DISCUSSIONS

This section evaluates the performance of various smart farming recommendations by focusing on four key data types: weather conditions, soil properties, fertilizer details, and crop information. These core elements significantly influence agricultural outcomes. The models were developed using Python on a standard Windows 10 setup equipped with an Intel Core i5 processor, 8 GB RAM, and a 256 GB SSD—sufficiently powerful to manage and process large volumes of data efficiently.

To enhance farmers' understanding of how decisions are made, the LIME tool is used to highlight the most influential factors in each prediction. This transparency makes it easier for farmers to follow the recommendations and apply them with confidence in real-world agricultural tasks.

A. SwiFT for CROP RECOMMENDATION

The SwiFT model was evaluated on a dataset containing key soil parameters—nitrogen, phosphorus, potassium, soil pH, temperature, humidity, and rainfall—illustrated in Figure 5, all of which are vital for accurate crop recommendation. The dataset was split into **80% training** and **20% testing** to evaluate model generalization. During training, the model was optimized using the **Adam optimizer** with a **learning rate of 0.001**, a **batch size of 32**, and trained for **30 epochs**. SwiFT demonstrated the ability to efficiently handle high-dimensional data while maintaining robust performance.

Index	N	P	K	Temperature	Humidity	pH	Rainfall	Label	Label Encode
0	90	42	43	20.879765	82.0027	6.502	202.93	rice	18
13	93	46	36	24.014987	82.0568	6.984	185.27	rice	18
14	94	50	37	25.665821	80.6638	6.948	209.58	rice	18
17	91	35	39	23.793919	80.4181	6.971	206.26	rice	18
20	89	45	36	21.305241	80.4747	6.441	185.49	rice	18

Figure 5. Dataset for Crop recommendation

Implementing the SwiFT model results in a multi-class confusion matrix that visualizes its classification performance across various crop categories. The intensity of each cell's color reflects the frequency of predictions, with darker shades indicating higher values. As shown in Figure 6 the confusion matrix demonstrates that the model achieves high accuracy, with most predictions correctly matching the true labels. The SwiFT model attains an overall accuracy of 98.75%, indicating that it is highly effective and well-suited for crop classification

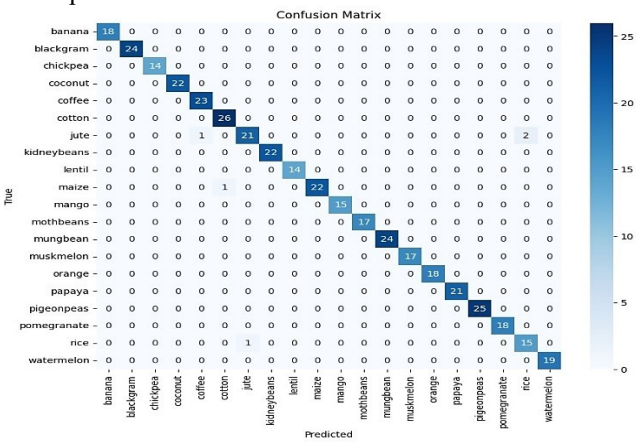

Figure 6. Confusion matrix of SwiFT

The classification report in Table 3 summarizes the performance of the SwiFT model used for crop recommendation. Most crops—such as banana, black gram, and chickpea—achieved perfect scores of 1.00 across precision, recall, and F1-score, indicating no misclassifications for those classes. However, a few crops showed minor misclassifications. Coffee, cotton, and maize had slightly lower precision or recall values (approximately 0.96–0.98). Jute had a recall of 0.88, indicating that the model failed to identify some jute instances. Similarly, rice had a precision of 0.88, suggesting a few incorrect predictions were made for that class.

Table 3. Classification report of crop recommendation

Class	Precision	Recall	F1-Score	Support
banana	1.00	1.00	1.00	18
black gram	1.00	1.00	1.00	24
chickpea	1.00	1.00	1.00	14
coconut	1.00	1.00	1.00	22
coffee	0.96	1.00	0.98	23
cotton	0.96	1.00	0.98	26
jute	0.95	0.88	0.91	24
kidney beans	1.00	1.00	1.00	22
lentil	1.00	1.00	1.00	14
maize	1.00	0.96	0.98	23
mango	1.00	1.00	1.00	15
moth beans	1.00	1.00	1.00	17
mung bean	1.00	1.00	1.00	24
muskmelon	1.00	1.00	1.00	17
orange	1.00	1.00	1.00	18
papaya	1.00	1.00	1.00	21
pigeon peas	1.00	1.00	1.00	25
pomegranate	1.00	1.00	1.00	18
rice	0.88	0.94	0.91	16
watermelon	1.00	1.00	1.00	19
Accuracy			0.99	400
macro avg	0.99	0.99	0.99	400
weighted avg	0.99	0.99	0.99	400

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By passing values of parameters (N, P, K, Ph, temp, humidity) the model tells the best crop for field which is predicted by model can see in Figure 7.

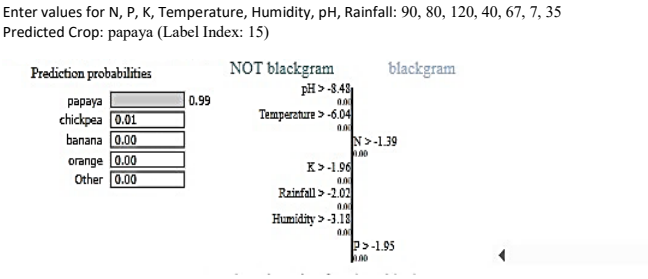

Figure 7. Testing of crop recommendation

A comparative performance analysis is shown in Figure 8 demonstrates different ML models (Random Forest, SVM, Logistic Regression, MLP, and SwiFT) for crop prediction, using metrics like accuracy and F1 score.

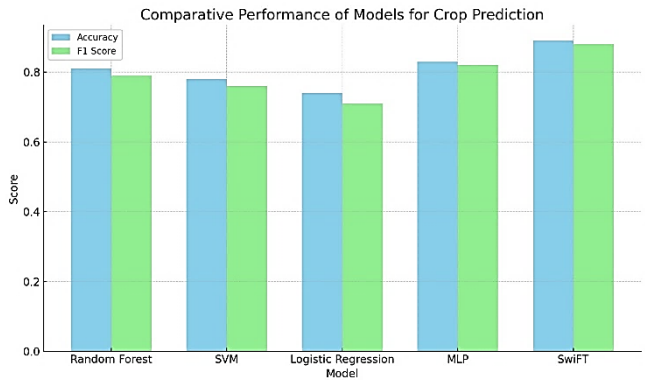

Figure 8. Comparison of SwiFT with other models

Ablation studies summarized in Figure 9 indicate that the crop recommendation model achieves its highest accuracy of 99% when all features—soil nutrients (N, P, K), temperature, humidity, pH, and rainfall—are used together.

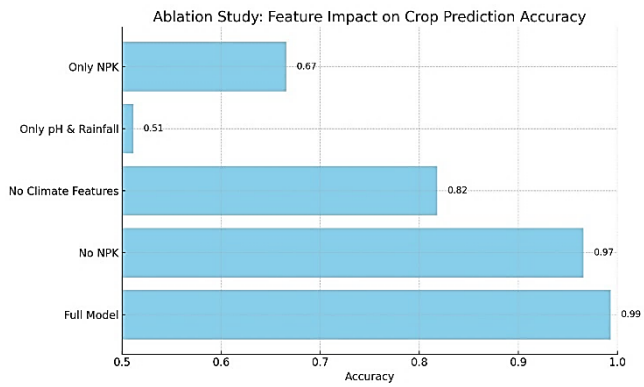

Figure 9. Ablation study of crop recommendation

When either soil nutrients or climate-related features are removed, the model's accuracy drops to 97% and 82%, respectively. Models using only NPK or only pH and rainfall perform significantly worse, with accuracies of 67% and 51%, respectively. These results highlight the importance of combining both soil and climatic features for optimal model performance in crop recommendation.

B. TTL for IRRIGATION ADVICE

The TTL model is used to predict irrigation needs based on essential environmental features such as soil moisture, temperature, air humidity, and current irrigation status. These parameters form the foundation for the model to determine whether a crop requires irrigation at a given time. The dataset provided in Figure 10, was split into 80% training and 20% testing to evaluate the model's generalization and accuracy. The model was trained using the Adam optimizer with a learning rate of 0.001 for 30 epochs, ensuring stable and efficient learning.

Index	Soil Moisture	Temperature	Air Humidity	Irrigation need	Irrigation Advice
0	683.801	29.184908	71.78	0	Sufficient Moisture (No Irrigation need)
1	409.571	33.707225	77.97	1	Moderate (Irrigation Recommended)
2	659.092	24.760311	60.77	1	Moderate (Irrigation Recommended)
3	842.929	32.738315	59.32	0	Sufficient Moisture (No Irrigation need)
4	414.199	25.692744	66.62	1	Moderate (Irrigation Recommended)

Figure 10. Dataset for irrigation advice

By implementing Transformer-base Tabular Learning model an accuracy of **99.13%** was achieved. As shown in Figure 11, the confusion matrix evaluates the performance of the TTL model for irrigation recommendation based on soil moisture levels. The results demonstrate strong model performance, with most predictions correctly matching the actual moisture levels.

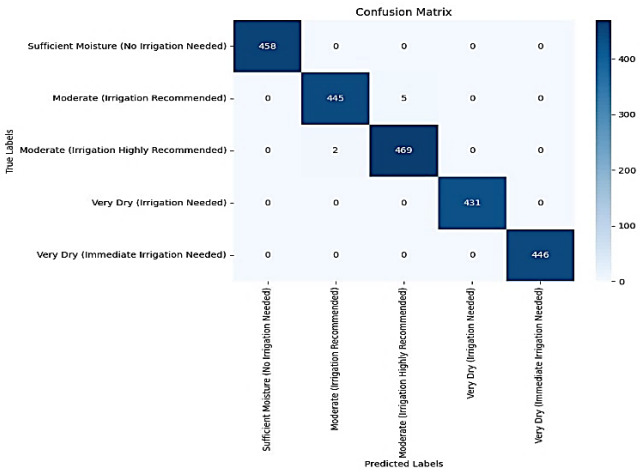

Figure 11. Confusion matrix of TTL

The classification report presented in Table 4 demonstrates the strong performance of the model, with overall accuracy, precision, recall, and F1-score all at approximately 0.99.

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Table 4. Classification report of Irrigation advice

Class	Precision	Recall	F1-Score	Support
Moderate (Irrigation Highly Recommended)	1.00	1.00	1.00	458
Moderate (Irrigation Recommended)	1.00	0.96	0.98	450
Sufficient Moisture (No Irrigation Needed)	0.97	1.00	0.98	471
Very Dry (Immediate Irrigation Needed)	0.98	0.99	0.98	427
Very Dry (Irrigation Needed)	0.99	0.98	0.99	450
Accuracy			0.99	2256
Macro Avg	0.99	0.99	0.99	2256
Weighted Avg	0.99	0.99	0.99	2256

This indicates that model provides highly accurate and consistent recommendations for irrigation needs across all tested scenarios.

By giving values of soil moisture, air humidity, temperature the model tells the dryness of soil and predict the advice whether there is an irrigation need or not can see in Figure 12.

```
# Example Usage

soil moisture = float (input ("Enter Soil Moisture: "))
temperature = float (input ("Enter Temperature: "))
air humidity = float (input ("Enter Air Humidity: "))

prediction = load_model_and_predict (soil moisture, temperature, air humidity)
print ("Irrigation Advice:", prediction)

Enter Soil Moisture: 200
Enter Temperature: 15
Enter Air Humidity: 34

Irrigation Advice: Very Dry (Irrigation Needed)
```

Figure 12. Testing of TTL on irrigation advice

Figure 13 presents the results of an ablation study, illustrating how the model's accuracy is affected by the removal of individual features.

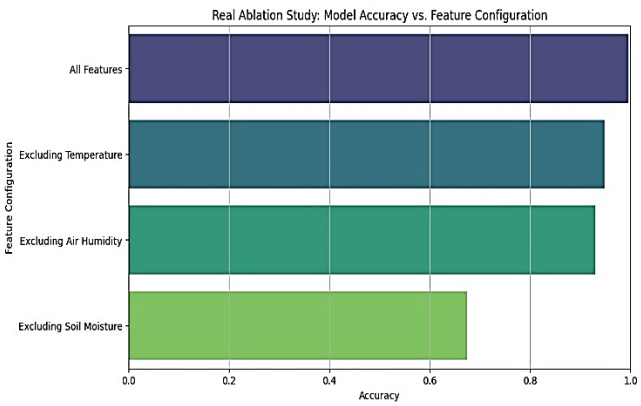

Figure 13. Ablation study of irrigation advice

Using all features yields the highest accuracy, while excluding any single feature leads to a noticeable drop in performance. Temperature and soil moisture appear to be the most critical features, as their exclusion results in the largest declines in accuracy. This study highlights the importance of careful feature selection to achieve optimal model performance.

C. TabNet regressor +XAI for SOIL ANALYSIS

TabNet combined with Explainable AI (XAI) can function effectively as a regressor for soil analysis, particularly when predicting continuous numerical values. These values may include soil nutrient levels, moisture content, organic matter percentage, or a fertility index. To assess soil health, key parameters such as nitrogen, phosphorus, potassium, pH, and moisture are essential. By evaluating these factors, the model can provide more accurate predictions of overall soil health. The features selected to train TabNet's regression model from the dataset are illustrated in Figure 14.

N	P	K	pH	Moisture	Soil Health
37.454012	51.908179	672.270299	7.196847	64.825695	61.084823
95.071431	47.918188	79.66814	6.818165	17.238636	70.478745
73.199394	2.564207	25.04679	5.79863	87.239456	40.885614
59.865848	34.124783	62.48741	4.52909	61.311624	49.638596
15.601864	38.019562	5.7174598	6.733076	15.720388	47.911425

Figure 14. Dataset for soil analysis

The TabNet regressor model achieves a Root Mean Squared Error (RMSE) of 1.51 and an R^2 of 98.72%. The low RMSE indicates that, on average, the difference between the actual and predicted soil health values is minimal. The high R^2 score signifies that 98.7% of the variance in soil health is explained by the model's predictions. This is considered excellent and demonstrates a very strong model fit, as illustrated in Figure 15.

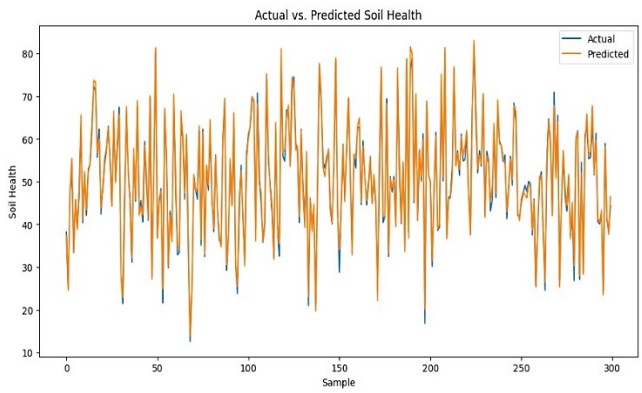

Figure 15. Actual vs Predicted soil analysis

By inputting the values of key parameters—nitrogen (N), phosphorus (P), potassium (K), pH, and moisture—the model evaluates whether the soil is in good condition or if it has any major issues. Figure 16 highlights that the model is well-trained and delivers appropriate and reliable results for soil health assessment.

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Figure 16. **TabNet+XAI on soil data for soil analysis**

An ablation study, presented in Figure 17, was conducted to evaluate the contribution of each feature to the model's performance.

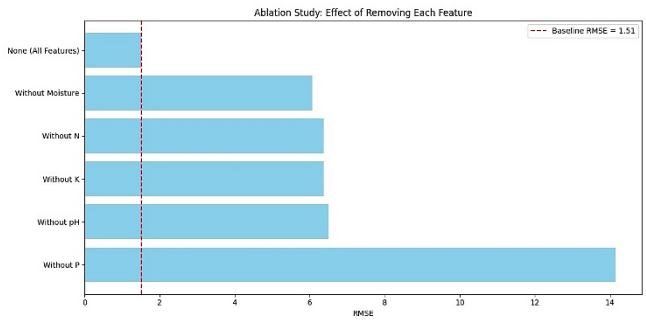

Figure 17. **Ablation study of soil analysis**

The results indicate that phosphorus (P) plays a critical role in prediction accuracy, as the RMSE increased significantly from 1.51 to over 14 when this feature was removed. Other features—moisture, nitrogen, potassium, and pH—also contribute substantially, with their exclusion resulting in RMSE values exceeding 6.0. These findings underscore the importance of maintaining a comprehensive soil profile, particularly phosphorus content, for achieving accurate soil analysis and effective recommendation systems.

D. TabNet classifier + XAI for fertilizer recommendation

TabNet combined with Explainable AI (XAI) is highly effective as a classifier for fertilizer recommendation. The goal is to categorize and suggest the most suitable fertilizer type based on soil and crop conditions. This approach offers both high accuracy and strong interpretability, making it ideal for real-world agricultural applications where informed decision-making is essential. The parameters used for optimal fertilizer recommendation including temperature, humidity, moisture, soil type, crop type, N, P, K, and fertilizer name are illustrated Figure 18.

Index	Temperature	Humidity	Moisture	Soil Type	Crop Type	N	K	P	Fertilizer Name
0	26	52	38	Sandy	Maize	37	0	0	Urea
1	29	52	45	Loamy	Sugarcane	12	0	36	DAP
2	34	65	62	Black	Cotton	7	9	30	14-35-14
3	32	62	34	Red	Tobacco	22	0	20	28-28
4	28	54	46	Clayey	Paddy	35	0	0	Urea

Figure 18. **Dataset for fertilizer recommendation**

Figure 19 highlights the performance of the TabNet classifier used for fertilizer recommendation. The confusion matrix shows an accuracy of 98.53%, F1-score of 98.47%, precision of 98.69%, and recall of 98.53%.

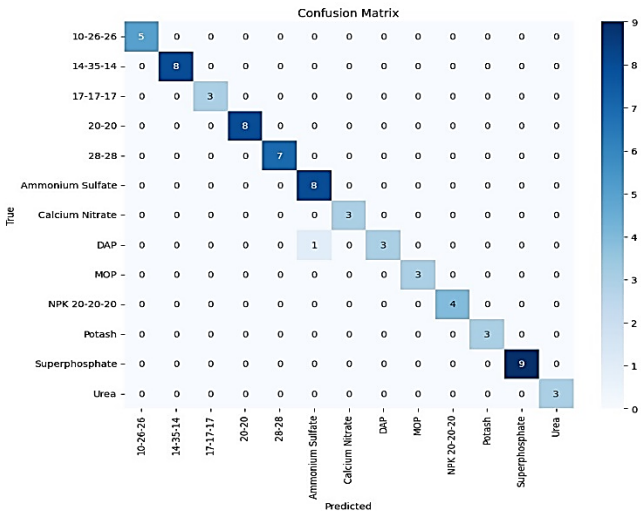

Figure 19. Confusion matrix of TabNet+XAI for fertilizer recommendation

These results confirm that the model is well-fitted and highly effective for making accurate fertilizer recommendations. The classification report provided in Table 5 shows that the fertilizer recommendation model performs exceptionally well, achieving an overall accuracy of 99%, along with high precision, recall, and F1-scores across nearly all fertilizer classes. Most classes are predicted perfectly, with only minor performance drops observed for classes 5 and 7. The macro and weighted averages, both at 99%, further confirm the model's strong and consistent performance, making it a highly reliable tool for fertilizer recommendation.

Table 5. **Classification report for fertilizer recommendation.**

Class	Precision	Recall	F1-Score	Support
0	1.00	1.00	1.00	5
1	1.00	1.00	1.00	8
2	1.00	1.00	1.00	3
3	1.00	1.00	1.00	8
4	1.00	1.00	1.00	7
5	0.89	1.00	0.94	8
6	1.00	1.00	1.00	3
7	1.00	0.75	0.86	4
8	1.00	1.00	1.00	3
9	1.00	1.00	1.00	4
10	1.00	1.00	1.00	3
11	1.00	1.00	1.00	9
12	1.00	1.00	1.00	3
Accuracy			0.99	68
Macro Avg	0.99	0.98	0.98	68
Weighted Avg	0.99	0.99	0.98	68

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The TabNet-based fertilizer recommendation system, illustrated in Figure 20, utilizes XAI techniques to enhance transparency. The LIME visualization provides interpretability by helping farmers understand the decision-making process behind fertilizer selection.

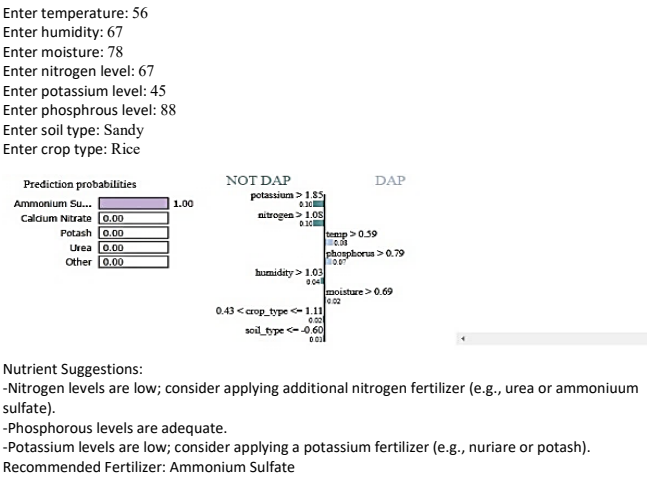

Figure 20. TabNet+XAI on fertilizer recommendation

Figure 21 illustrates an AI-powered recommendation system designed to be simple and farmer-friendly. Whether farmers connect IoT sensors or manually enter key details such as soil and weather conditions, the system quickly provides helpful suggestions for crop selection, irrigation planning, fertilizer recommendation, and soil health assessment. The user interface is clean and intuitive, making it easy to input data and receive personalized, actionable guidance. This enables farmers to make smart, data-driven decisions, helping them grow healthier crops and manage their farms more efficiently.

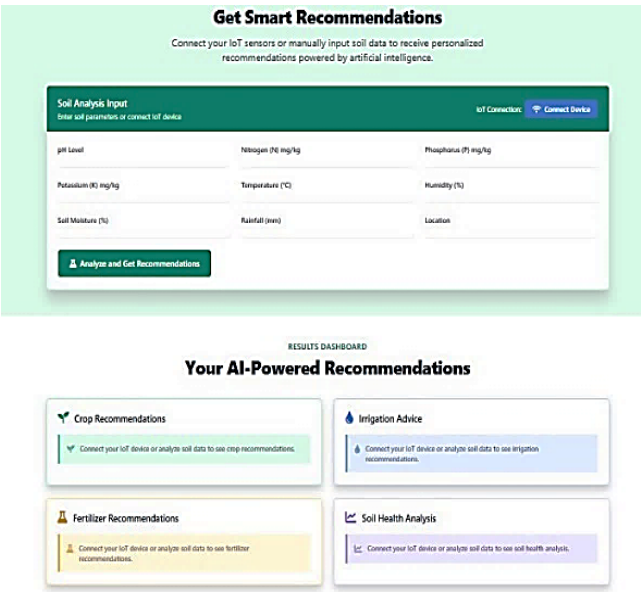

Figure 21. User interface of proposed model

Implementing this AI-driven agriculture system in real-world settings is not as straightforward as flipping a switch—several practical challenges must be considered. First, what works well in one region may not perform the same in another due to variations in soil types, climatic conditions, and crop varieties, requiring model adaptation and fine-tuning. Maintaining IoT sensors, especially in remote or rural areas, can also be difficult, and sensor failures can significantly reduce the usefulness of the collected data.
Additionally, limited internet connectivity remains a barrier for many farmers, making it challenging to access real-time updates. For widespread adoption, the system must be simple, affordable, and available in local languages, particularly for farmers who may have limited exposure to technology. Data privacy and equitable access are also key concerns—the system should ensure that all farmers, regardless of scale, benefit equally. In summary, for the system to deliver real value on the ground, it must be reliable, user-friendly, and adaptable to diverse farming conditions.

Limitations and Future Directions: Despite the strong potential of the proposed AI-driven farming system, several real-world challenges remain to be addressed. A key limitation is the durability and maintenance of IoT sensors, particularly in outdoor farming environments where dust, heat, and moisture can impact their performance. Additionally, the cost of deploying sensors and associated infrastructure may hinder adoption, especially among resource-constrained farmers. While the system has demonstrated promising results with the current datasets, its performance must be validated across diverse soil profiles, climatic conditions, and crop types to ensure broad applicability and reliability. Looking ahead, one promising improvement is the integration of transfer learning techniques, which would allow the models to adapt to new regions with minimal retraining. Another valuable future direction involves developing a mobile application that enables farmers to easily input field data, receive real-time recommendations, and visualize insights on crop and soil health. Such tools would make smart farming technologies more accessible, especially in rural areas with limited digital infrastructure, thereby increasing the impact and reach of AI-driven agricultural solutions.

V. CONCLUSION

The integration of IoT, transformer-based models, and deep learning in agriculture has demonstrated significant potential, transforming traditional farming into a smarter, more efficient, and environmentally sustainable practice. The system developed in this project offers a comprehensive solution by unifying soil analysis, irrigation planning, crop recommendation, and fertilizer selection. These components address critical agricultural challenges such as resource limitations, climate change, and declining soil health, thereby supporting a more sustainable and productive future for farming.
The system's performance is validated through the strong results delivered by its individual modules:

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- The **TTL model** provided irrigation advice with an impressive **accuracy of 99.13%**, ensuring efficient water usage.
- The **SwiFT model** achieved **98.75% accuracy** in crop recommendation, supporting optimal crop selection.
- The **TabNet regressor**, enhanced with **XAI**, analyzed soil health with an **R^2 score of 98.7%**, making it highly effective for assessing nutrient levels and overall soil conditions.
- The **TabNet classifier** reached **99.3% accuracy** in fertilizer recommendation, enabling precise and informed decision-making.

Designed with a high degree of modularity, the proposed system leverages real-time data processing for advanced analytics and employs edge computing and cloud-based AI, ensuring scalability across diverse agricultural environments. Furthermore, the use of federated learning supports collaborative model training while preserving data privacy.

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